

EXISTS & NOT EXISTS Operators

Mastering Correlated Subqueries & Execution Flow for Competitive Exams

1. Core Concepts

EXISTS

- Used primarily with **correlated subqueries**.
- Evaluates whether the inner query returns **at least one matching row**.
- Returns a Boolean result (TRUE or FALSE).
- Stops searching as soon as a single match is found (Short-circuiting).

NOT EXISTS

- Evaluates to TRUE if the inner subquery returns **zero rows**.
- Used to find outer records that have **no corresponding matches** in the inner table.
- Returns FALSE as soon as a single match is encountered.

2. SQL Code Examples & Semantics

Example A: Finding employees assigned to at least one project

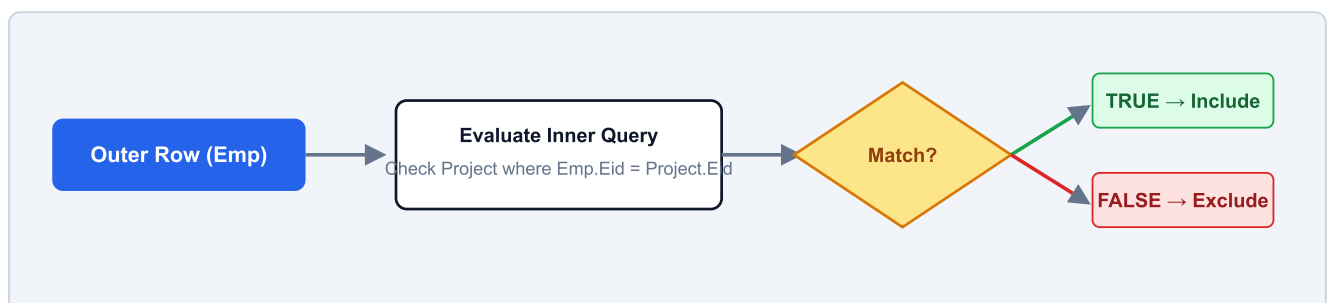
```
SELECT * FROM Emp WHERE EXISTS ( SELECT Eid FROM Project WHERE Emp.Eid = Project.Eid )
```

Example B: Finding employees working on zero projects

```
SELECT * FROM Emp WHERE NOT EXISTS ( SELECT Eid FROM Project WHERE Emp.Eid = Project.Eid )
```

3. Execution Mechanism: How EXISTS Works

Unlike IN (where the inner query completes first), EXISTS runs in a **top-to-bottom, row-by-row correlated evaluation** flow:



4. Key Exam Comparison: IN vs. EXISTS